Kernix 0.12.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

46 workloads

156 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

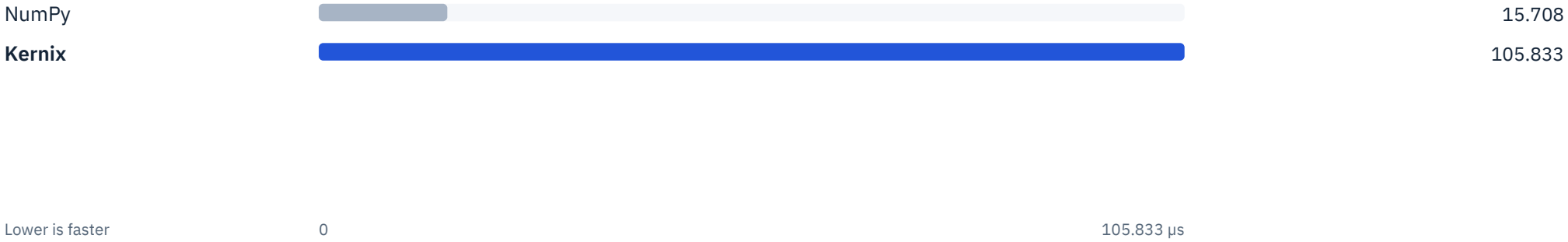
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication · transposed view

Per-call median

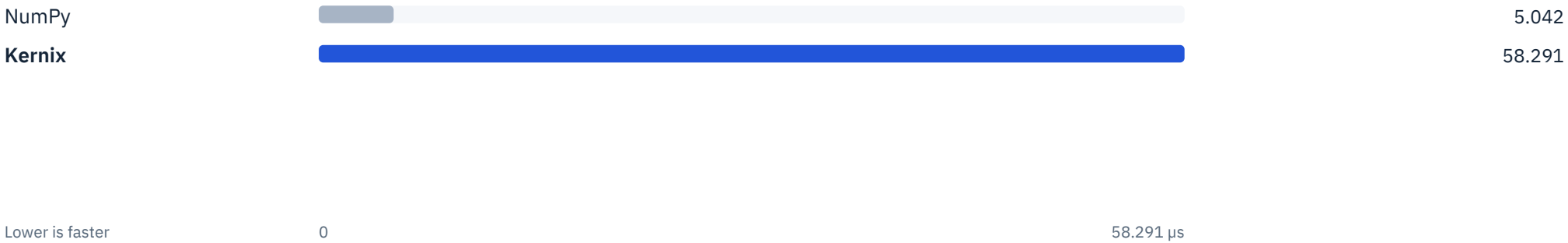
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



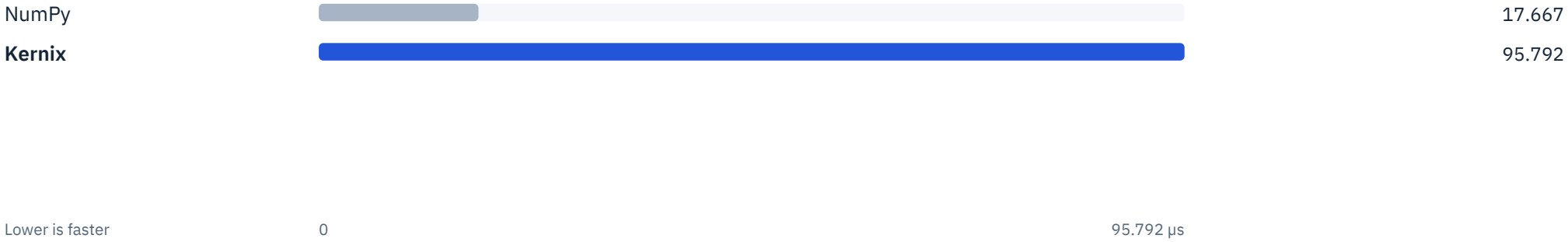
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

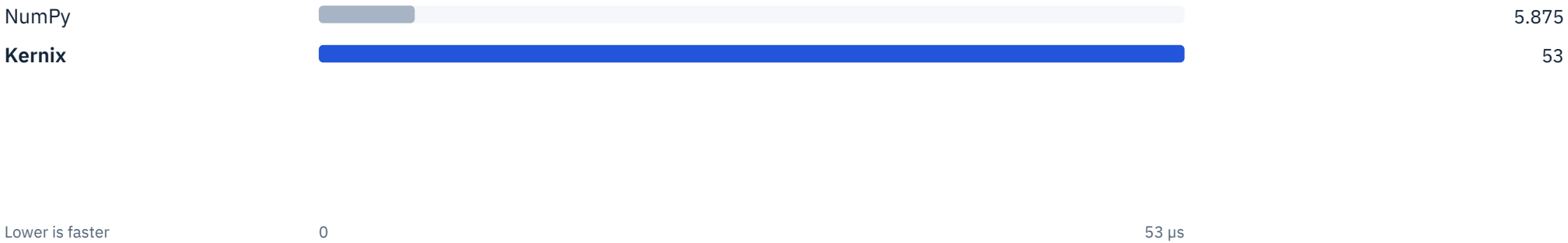
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



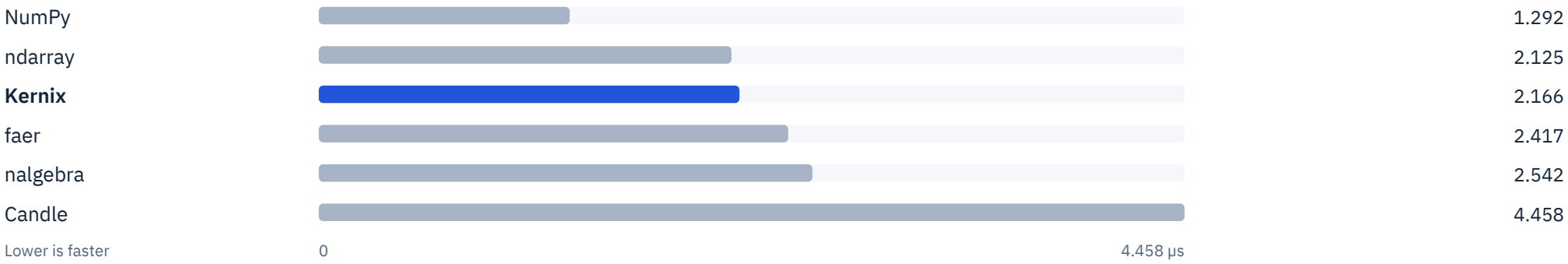
Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

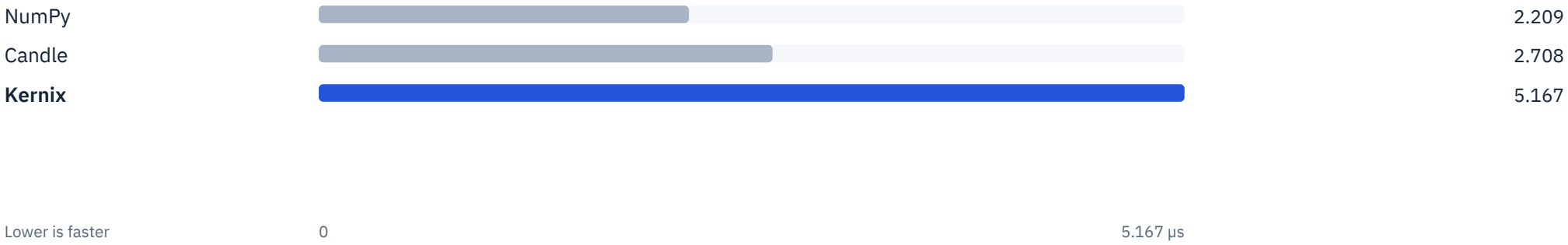
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



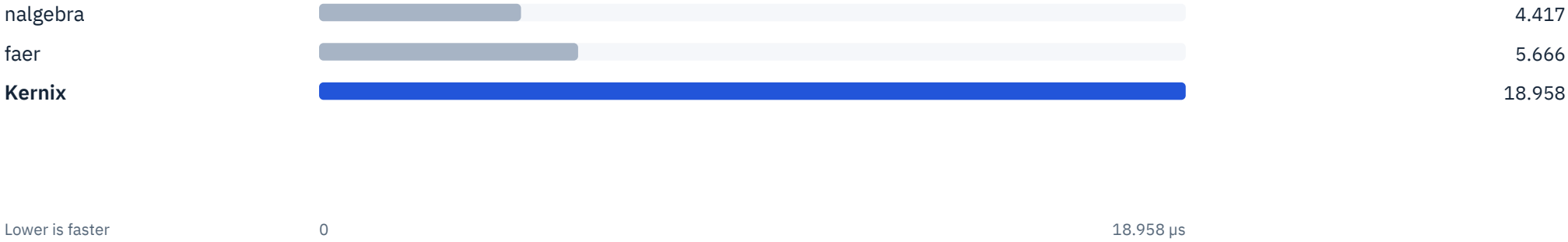
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



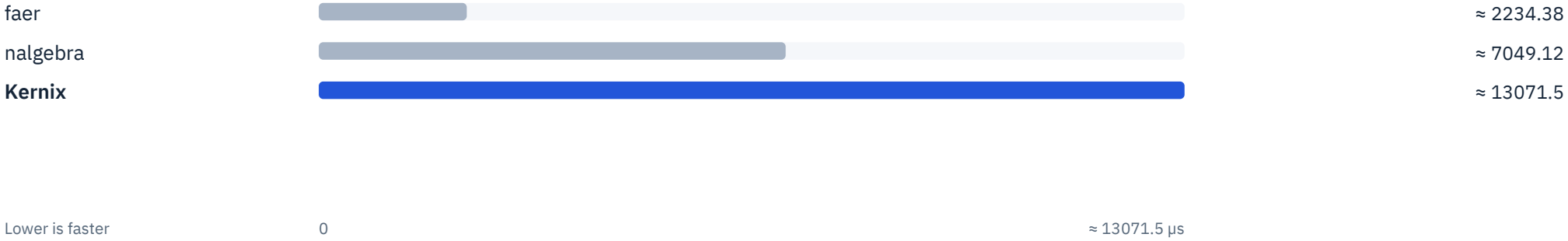
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



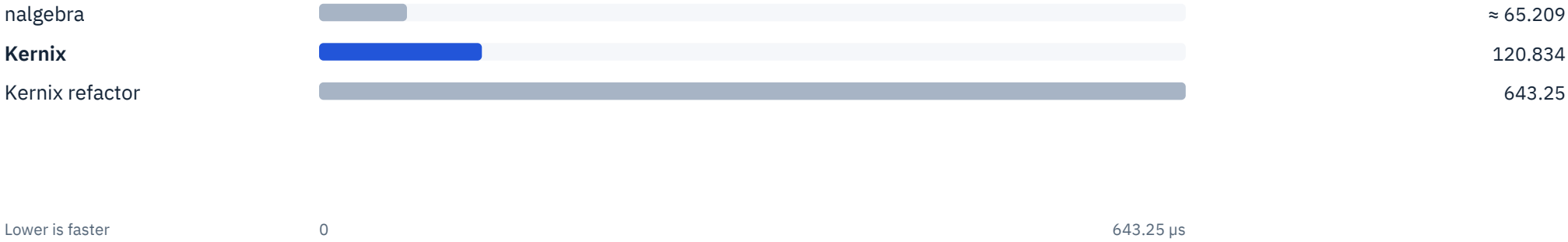
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

Per-call median

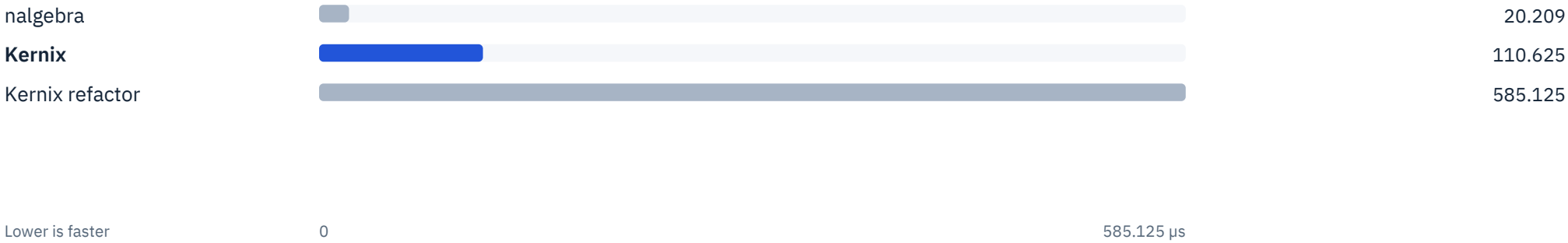
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



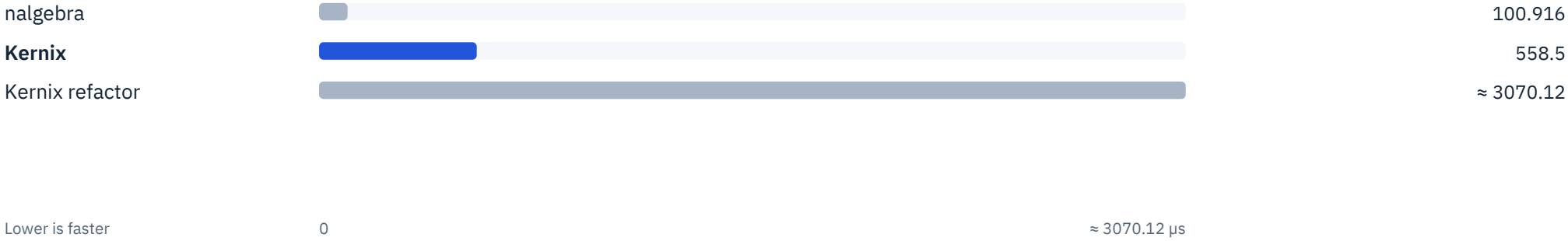
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median



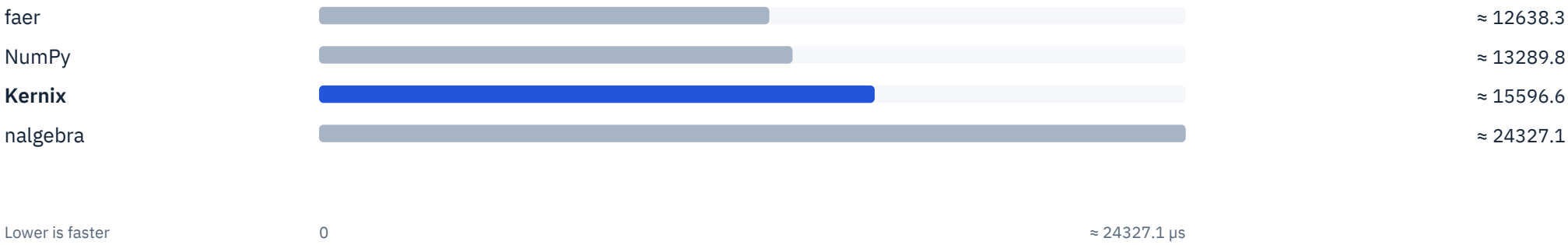
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

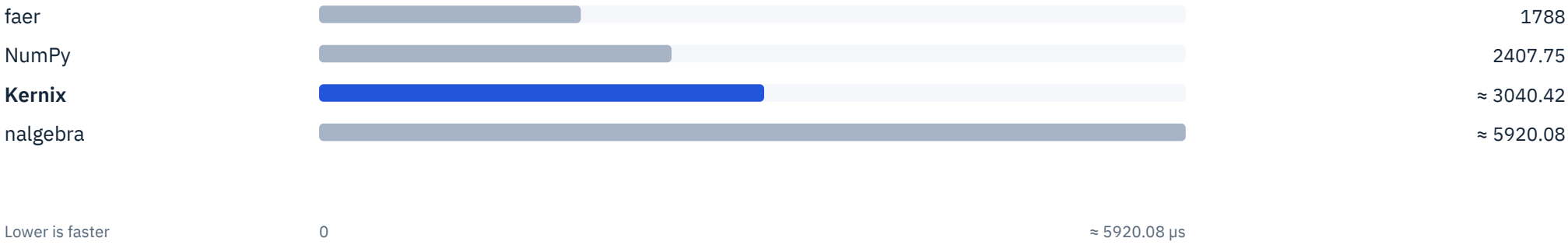
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



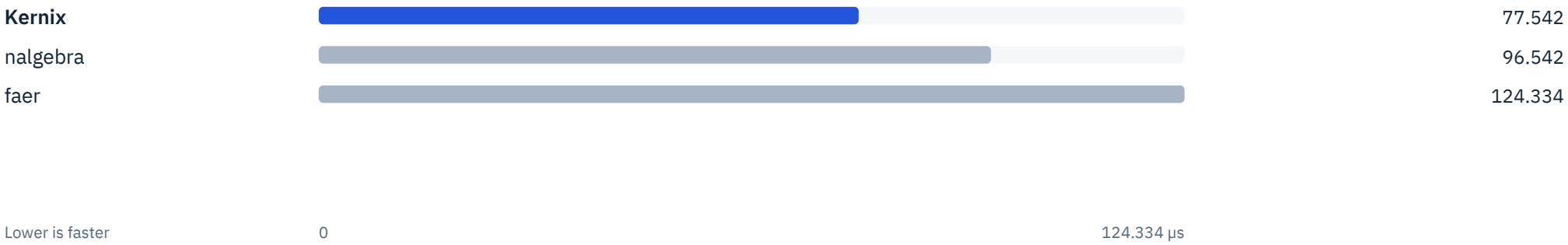
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

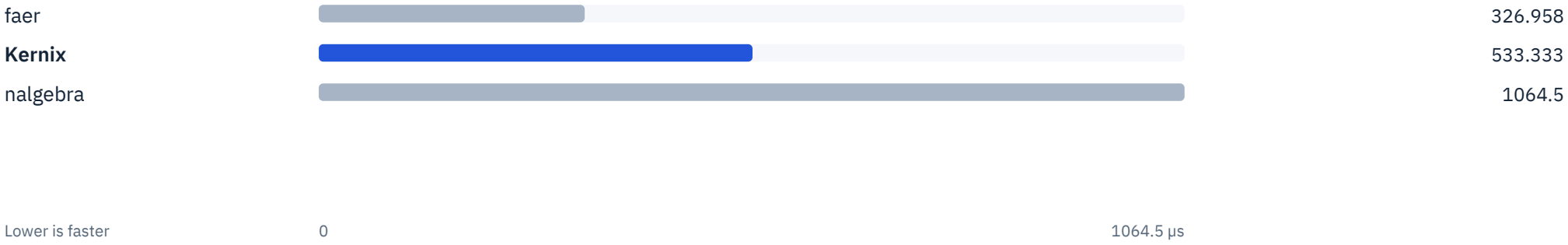
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

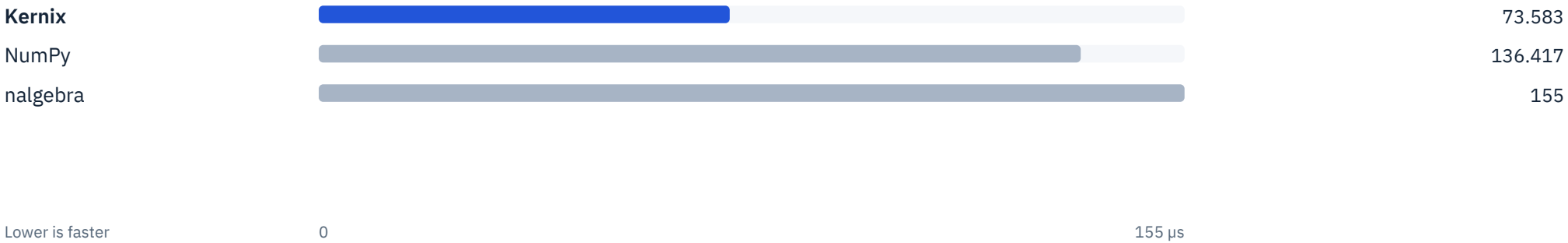
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

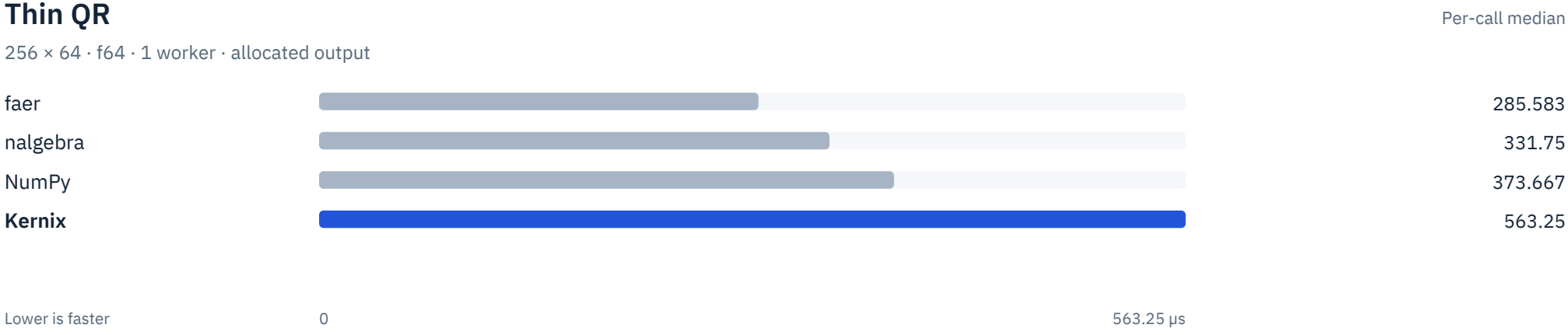
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



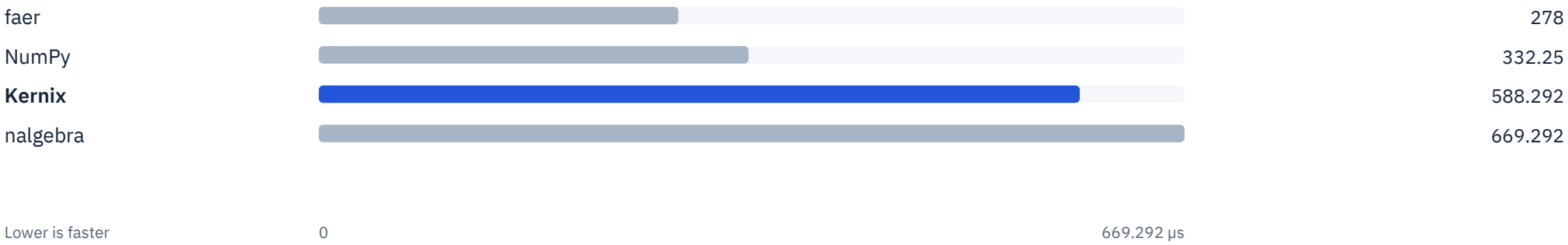
Kernix 0.12.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

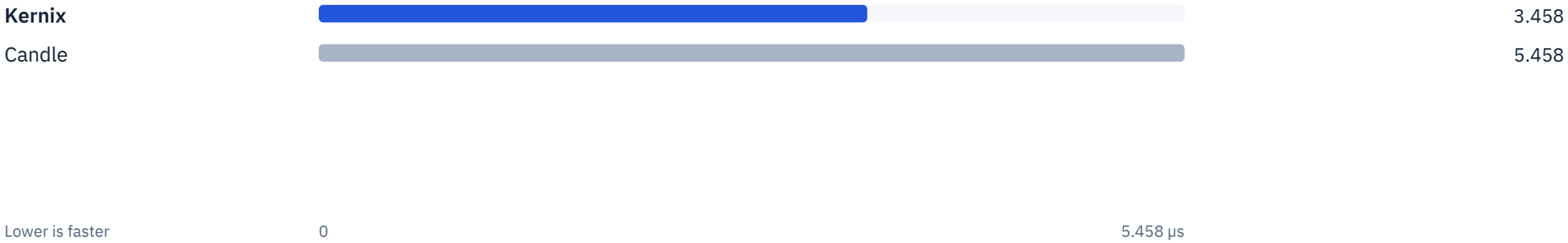
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

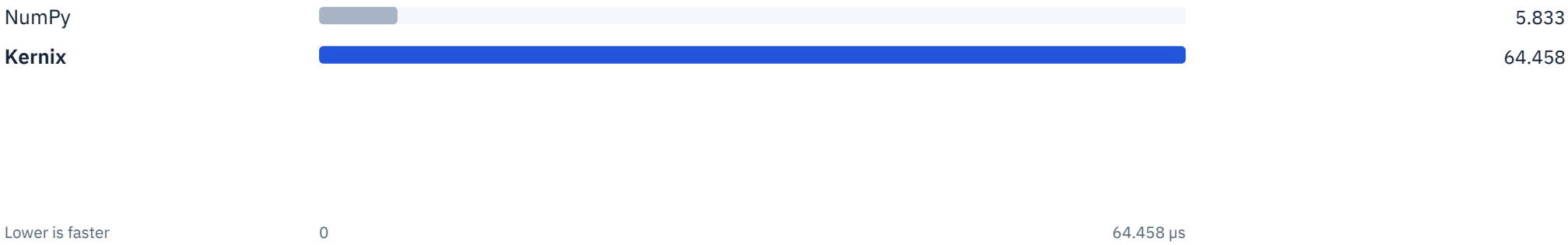


Kernix 0.12.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



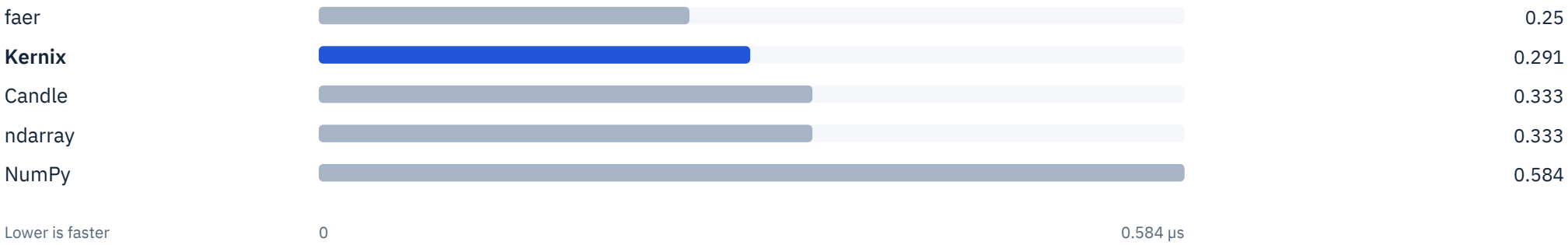
Kernix 0.12.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

64 × 64 · f32 · 1 worker · allocated output

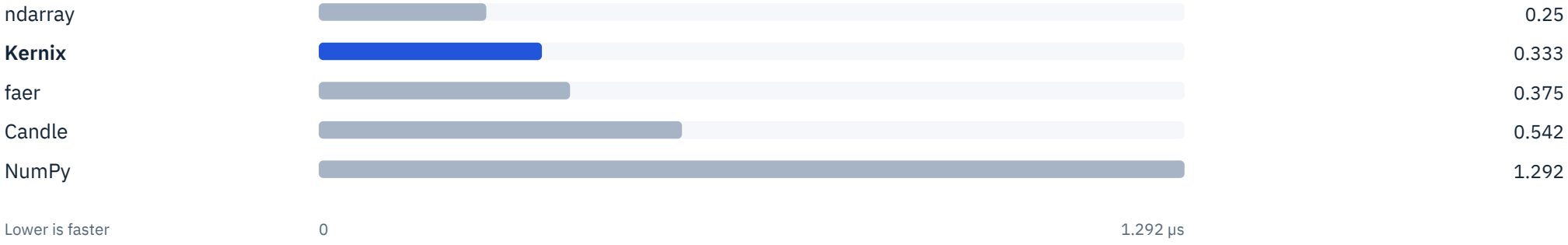
Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9999363422242026	0.8434728164457894 – 1.0342578520770012

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.3391463976714771	0.3347444354363207 – 0.34212995784873657

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

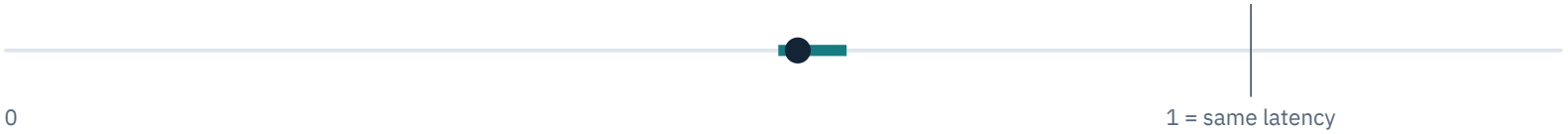


CURRENT / BASELINE	95% INTERVAL
0.3395659435871721	0.29953666777081356 – 0.3520476746781338

Thin QR

512 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.6364933793072876	0.6209107889725535 – 0.6756612892776158

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.5251560195707211	0.4995823827409169 – 0.533394334068534

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0065869156092198	0.9334875283446712 – 1.0366644664466447

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.9692970033422081 – 1.0242323200992556

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	1.0203252032520325	95% INTERVAL	0.9828453796889296 – 1.0506988647316935

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	**1**	95% INTERVAL	0.9642077695329551 – 1.0186666666666666

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	**1.0341187136270142**	95% INTERVAL	0.9979659367396594 – 1.0490804116484436

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9852494088503547	0.9299589318823753 – 1.0053293424102498

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9968545275313336	0.9725039674746369 – 1.0434932624259767

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9854261083743843	0.9696438041145655 – 0.995589617708063

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.98656	0.9482543640897756 – 1.1184085885696242

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9959645031113004	0.9219876288659794 – 1.0164903225806452

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0047169811320755	0.9952867538931413 – 1.039894165994258

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.8290898448931812 – 1.0305454545454547

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	1.0229132569558101	95% INTERVAL	0.9333333333333333 – 1.0463169642857142

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.995259977566306	0.9373491217847332 – 1.0350073578595318

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9960448117608732	0.9707239802826404 – 1.0095877952453927

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9951801061606067	0.9423570914820015 – 1.0506741840165743

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0216484838542095	0.5988323603002502 – 1.2320360685302074

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0137858548492327	0.7405825022956959 – 1.0541735305919873

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0029569620253165	0.9843949367088608 – 2.0068221741084193

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	105833	105.833
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	15708	15.708
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	58291	58.291
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	5042	5.042
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	95792	95.792
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	17667	17.667
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	5875	5.875
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	53000	53
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2542	2.542
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	4458	4.458

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1292	1.292
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2125	2.125
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2166	2.166
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3500	3.5
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	14333	14.333
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14833	14.833
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13750	13.75
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	12625	12.625
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13959	13.959

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	102000	102
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	96667	96.667
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	107042	107.042
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	107083	107.083
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	94083	94.083
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	17875	17.875
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	17708	17.708
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	876167	876.167
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	117000	117
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	815042	815.042

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	797041	797.041
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	741292	741.292
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	123583.99999999999	123.58399999999999
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	893792	893.792
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7708	7.708
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1917	1.917
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7334	7.334
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6917	6.917
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7750	7.75
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7083	7.083

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	5834	5.834
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	49542	49.542
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	5917	5.917
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	51125	51.125
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	48916	48.916
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	51541	51.541
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	50875	50.875
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	438250	438.25
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	436083	436.083
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	35291	35.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	365334	365.334
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	366667	366.667
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	35625	35.625
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	428916	428.916
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2084	2.084
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2958	2.958
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3750	3.75
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	5167	5.167
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2209	2.209
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2708	2.708

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	50583	50.583
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	152791	152.791
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	1184417	1184.417
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	47500	47.5
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5666	5.666
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	4417	4.417
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18958	18.958
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1252875	1252.875
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	930500	930.5
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	297292	297.292

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	13071542	13071.542
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2234375	2234.375
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	7049125	7049.125
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	951000	951
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1370666	1370.666
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	322584	322.584
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	643250	643.25
Cholesky rank-one update / nalgebra 256 × 256 · f64	65209.00000000001	65.20900000000001
Cholesky rank-one update / Kernix 256 × 256 · f64	120834	120.834
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	20209	20.209

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	585125	585.125
Cholesky rank-one downdate / Kernix 256 × 256 · f64	110625	110.625
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3070125	3070.125
Cholesky rank-one update / nalgebra 512 × 512 · f64	100916	100.916
Cholesky rank-one update / Kernix 512 × 512 · f64	558500	558.5
Thin QR / Kernix 512 × 128 · f64	2764000	2764
Thin QR / nalgebra 512 × 128 · f64	2897042	2897.042
Thin QR / NumPy 512 × 128 · f64	2239542	2239.542
Thin QR / faer 512 × 128 · f64	1656917	1656.917
Thin QR / faer 1024 × 256 · f64	12638334	12638.334

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 1024 × 256 · f64	13289792	13289.792
Thin QR / nalgebra 1024 × 256 · f64	24327084	24327.084
Thin QR / Kernix 1024 × 256 · f64	15596583	15596.583
Cholesky factorization / nalgebra 512 × 512 · f64	5920084	5920.084
Cholesky factorization / faer 512 × 512 · f64	1788000	1788
Cholesky factorization / NumPy 512 × 512 · f64	2407750	2407.75
Cholesky factorization / Kernix 512 × 512 · f64	3040417	3040.417
Column-pivoted QR / Kernix 96 × 32 · f64	77542	77.542
Column-pivoted QR / faer 96 × 32 · f64	124334	124.334
Column-pivoted QR / nalgebra 96 × 32 · f64	96542	96.542

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 256 × 64 · f64	1064500	1064.5
Column-pivoted QR / faer 256 × 64 · f64	326958	326.958
Column-pivoted QR / Kernix 256 × 64 · f64	533333	533.333
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	72458	72.458
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	79542	79.542
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	54375	54.375
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	73583	73.583
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	136417	136.417
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	155000	155
Thin QR / NumPy 96 × 32 · f64	46958	46.958

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	50208	50.208
Thin QR / nalgebra 96 × 32 · f64	45250	45.25
Thin QR / Kernix 96 × 32 · f64	55333	55.333
Thin QR / NumPy 256 × 64 · f64	373667	373.667
Thin QR / nalgebra 256 × 64 · f64	331750	331.75
Thin QR / Kernix 256 × 64 · f64	563250	563.25
Thin QR / faer 256 × 64 · f64	285583	285.583
Cholesky factorization / faer 64 × 64 · f64	10083	10.083
Cholesky factorization / Kernix 64 × 64 · f64	20208	20.208
Cholesky factorization / NumPy 64 × 64 · f64	13750	13.75

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 64 × 64 · f64	19500	19.5
Cholesky factorization / faer 128 × 128 · f64	51542	51.542
Cholesky factorization / NumPy 128 × 128 · f64	47125	47.125
Cholesky factorization / nalgebra 128 × 128 · f64	117000	117
Cholesky factorization / Kernix 128 × 128 · f64	165833	165.833
Cholesky factorization / nalgebra 256 × 256 · f64	669292	669.292
Cholesky factorization / Kernix 256 × 256 · f64	588292	588.292
Cholesky factorization / NumPy 256 × 256 · f64	332250	332.25
Cholesky factorization / faer 256 × 256 · f64	278000	278
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4541	4.541

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6375	6.375
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3000	3
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6583	6.583
Model inference / Candle batched normalized/sliced MLP · f64	6458	6.458
Model inference / Kernix batched normalized/sliced MLP · f64	2084	2.084
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3458	3.458
Model recording + forward / Candle batched normalized/sliced MLP · f64	5458	5.458
Model backward / Kernix prepared graph → fresh gradients · f64	3667	3.667
Model backward / Candle prepared graph → fresh gradients · f64	19125	19.125
Parameter update / Kernix transactional SGD, fixed gradients · f64	750	0.75

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	64458	64.458
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5833	5.833
Row softmax / Candle 256 × 64 · f64	75625	75.625
Row softmax / Kernix 256 × 64 · f64	84541	84.541
Row softmax / ndarray 256 × 64 · f64	66000	66
Row softmax / NumPy 256 × 64 · f64	68958	68.958
Create zeros / ndarray 64 × 64 · f32	333	0.333
Create zeros / NumPy 64 × 64 · f32	584	0.584
Create zeros / Kernix 64 × 64 · f32	291	0.291
Create zeros / faer 64 × 64 · f32	250	0.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	333	0.333
Create ones / Candle 64 × 64 · f32	542	0.542
Create ones / NumPy 64 × 64 · f32	1292	1.292
Create ones / faer 64 × 64 · f32	375	0.375
Create ones / ndarray 64 × 64 · f32	250	0.25
Create ones / Kernix 64 × 64 · f32	333	0.333